Proof trees

We use these proof trees to determine or check the type of a term, or find an appropriate term for a given type. In order to do the latter, a placeholder must be used for the term (such as a long empty space, rectangles, various underlines, or variables not used for anything else, such as t, u, v, X, Y, etc...), and this must be continually refined as the applicable rule is selected and applied. Keep in mind that any refinement on the term must be propagated *downwards* in the proof tree.

Rules

The rules are organised into trees of their own, which may be constructed from the formal description of a language by taking the conclusion of each term constructor as the conclusion of the rule, and all the conditions separated by arrows – if any – as a separate condition for the rule. Higher-order functions are treated in a special way. See below for examples.

Sequent calculus

The logic we are using is similar to a single-conclusion sequent calculus. A $\vdash$ symbol separates a (possibly empty) context from the term we are examining. When the context is empty, the $\vdash$ symbol is omitted. The contexts can usually be arbitrarily extended (i.e. most rules and proof trees should include an arbitrary set Γ in the context, which we omit for brevity), enabling us to use the rules with any context. Contexts must be carried *upwards* in the proof tree. Several rules of sequent calculus may be used in these proofs, but the only one really needed is the following, often called `hyp`, for hypothesis.

$$\frac{}{\Gamma,\ A,\ \Delta \vdash A}$$

Razor

Below are the usual rules for the Razor language.

$$\text{true} : \text{Tm}\ \mathit{Bool} \qquad \frac{}{\text{true} : \text{Tm}\ \mathit{Bool}}$$

$$\text{false} : \text{Tm}\ \mathit{Bool} \qquad \frac{}{\text{false} : \text{Tm}\ \mathit{Bool}}$$

$$\text{ite} : \forall A.\ \text{Tm}\ \mathit{Bool} \to \text{Tm}\ A \to \text{Tm}\ A \to \text{Tm}\ A \qquad \frac{t : \text{Tm}\ \mathit{Bool} \qquad u : \text{Tm}\ A \qquad v : \text{Tm}\ A}{\text{ite}\ t\ u\ v : \text{Tm}\ A}$$

$$\text{num} : \mathbb{N} \to \text{Tm}\ \mathit{Nat} \qquad \frac{}{\text{num}\ n : \text{Tm}\ \mathit{Nat}}$$

$$\text{isZero} : \text{Tm}\ \mathit{Nat} \to \text{Tm}\ \mathit{Bool} \qquad \frac{t : \text{Tm}\ \mathit{Nat}}{\text{isZero}\ t : \text{Tm}\ \mathit{Bool}}$$

$$_ + _ : \text{Tm}\ \mathit{Nat} \to \text{Tm}\ \mathit{Nat} \to \text{Tm}\ \mathit{Nat} \qquad \frac{t : \text{Tm}\ \mathit{Nat} \qquad u : \text{Tm}\ \mathit{Nat}}{t + u : \text{Tm}\ \mathit{Nat}}$$

STLC

We extend these rules with ones for **s**imply **t**yped **l**ambda **c**alculus. Note that the second-order parameter of lam modifies the context. When considering these rules, remember that $\Rightarrow$ is right associative, while $_ \cdot _$ is left associative. Additional bracketing might be required during the construction of the proof trees.

$$_ \cdot _ : \forall A\ B.\ \text{Tm}\ (A \Rightarrow B) \to \text{Tm}\ A \to \text{Tm}\ B \qquad \frac{t : \text{Tm}\ (A \Rightarrow B) \qquad u : \text{Tm}\ A}{t \cdot u : \text{Tm}\ B}$$

$$\text{lam} : \forall A\ B.\ (\text{Tm}\ A \to \text{Tm}\ B) \to \text{Tm}\ (A \Rightarrow B) \qquad \frac{x : \text{Tm}\ A \vdash t : \text{Tm}\ B}{\text{lam}\ x.t : \text{Tm}\ (A \Rightarrow B)}$$

Examples

The following are example proof trees, showcased on the practices. I prefer to end finished branches with a ✓ to show that they are completed and I didn't just forget to finish them. These are optional. You are encouraged to derive these exercises again from scratch as practice. Note that in the first two examples, the created term is not the simplest possible one, rather it's what the exercise wanted.

`add2`

$$\dfrac{\dfrac{}{x : \mathrm{Tm}\ \mathit{Nat} \vdash x : \mathrm{Tm}\ \mathit{Nat}}\checkmark \quad \dfrac{}{x : \mathrm{Tm}\ \mathit{Nat} \vdash \mathrm{num}\ 2 : \mathrm{Tm}\ \mathit{Nat}}\checkmark}{\dfrac{x : \mathrm{Tm}\ \mathit{Nat} \vdash x + \mathrm{num}\ 2 : \mathrm{Tm}\ \mathit{Nat}}{\mathrm{lam}\ x. x + num2 : \mathrm{Tm}\ (\mathit{Nat} \Rightarrow \mathit{Nat})}}$$

`apply2x`

$$\dfrac{\dfrac{\dfrac{}{f : \mathrm{Tm}\ (\mathit{Nat} \Rightarrow \mathit{Nat}),\ x : \mathrm{Tm}\ \mathit{Nat} \vdash f : \mathrm{Tm}\ (\mathit{Nat} \Rightarrow \mathit{Nat})}\checkmark \quad \dfrac{}{f : \mathrm{Tm}\ (\mathit{Nat} \Rightarrow \mathit{Nat}),\ x : \mathrm{Tm}\ \mathit{Nat} \vdash x : \mathrm{Tm}\ \mathit{Nat}}\checkmark}{f : \mathrm{Tm}\ (\mathit{Nat} \Rightarrow \mathit{Nat}),\ x : \mathrm{Tm}\ \mathit{Nat} \vdash f \cdot x : \mathrm{Tm}\ \mathit{Nat}}}{\dfrac{f : \mathrm{Tm}\ (\mathit{Nat} \Rightarrow \mathit{Nat}),\ x : \mathrm{Tm}\ \mathit{Nat} \vdash f \cdot (f \cdot x) : \mathrm{Tm}\ \mathit{Nat}}{\dfrac{f : \mathrm{Tm}\ (\mathit{Nat} \Rightarrow \mathit{Nat}) \vdash \mathrm{lam}\ x. f \cdot (f \cdot x) : \mathrm{Tm}\ (\mathit{Nat} \Rightarrow \mathit{Nat})}{\mathrm{lam}\ f. \mathrm{lam}\ x. f \cdot (f \cdot x) : \mathrm{Tm}\ ((\mathit{Nat} \Rightarrow \mathit{Nat}) \Rightarrow \mathit{Nat} \Rightarrow \mathit{Nat})}}}$$

`K`

$$\dfrac{\dfrac{\dfrac{}{x : \mathrm{Tm}\ A,\ y : \mathrm{Tm}\ B \vdash x : \mathrm{Tm}\ A}\checkmark}{x : \mathrm{Tm}\ A \vdash \mathrm{lam}\ y. x : \mathrm{Tm}\ (B \Rightarrow A)}}{\mathrm{lam}\ x. \mathrm{lam}\ y. x : \mathrm{Tm}\ (A \Rightarrow B \Rightarrow A)}$$

`join`

$$\dfrac{\dfrac{\dfrac{\dfrac{}{f : \mathrm{Tm}\ (A \Rightarrow A \Rightarrow B),\ x : \mathrm{Tm}\ A \vdash f : \mathrm{Tm}\ (A \Rightarrow A \Rightarrow B)}\checkmark \quad \dfrac{}{f : \mathrm{Tm}\ (A \Rightarrow A \Rightarrow B),\ x : \mathrm{Tm}\ A \vdash x : \mathrm{Tm}\ A}\checkmark}{f : \mathrm{Tm}\ (A \Rightarrow A \Rightarrow B),\ x : \mathrm{Tm}\ A \vdash f \cdot x : \mathrm{Tm}\ (A \Rightarrow B)} \quad \dfrac{}{f : \mathrm{Tm}\ (A \Rightarrow A \Rightarrow B),\ x : \mathrm{Tm}\ A \vdash x : \mathrm{Tm}\ A}\checkmark}{f : \mathrm{Tm}\ (A \Rightarrow A \Rightarrow B),\ x : \mathrm{Tm}\ A \vdash f \cdot x \cdot x : \mathrm{Tm}\ B}}{\dfrac{f : \mathrm{Tm}\ (A \Rightarrow A \Rightarrow B) \vdash \mathrm{lam}\ y. f \cdot x \cdot x : \mathrm{Tm}\ (A \Rightarrow B)}{\mathrm{lam}\ f. \mathrm{lam}\ x. f \cdot x \cdot x : \mathrm{Tm}\ ((A \Rightarrow A \Rightarrow B) \Rightarrow A \Rightarrow B)}}$$

`S`

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{x : \mathrm{Tm}\ (A \Rightarrow B \Rightarrow C),\ y : \mathrm{Tm}\ (A \Rightarrow B),\ z : \mathrm{Tm}\ A \vdash x : \mathrm{Tm}\ (A \Rightarrow B \Rightarrow C)}\checkmark \quad \dfrac{}{x : \mathrm{Tm}\ (A \Rightarrow B \Rightarrow C),\ y : \mathrm{Tm}\ (A \Rightarrow B),\ z : \mathrm{Tm}\ A \vdash z : \mathrm{Tm}\ A}\checkmark}{x : \mathrm{Tm}\ (A \Rightarrow B \Rightarrow C),\ y : \mathrm{Tm}\ (A \Rightarrow B),\ z : \mathrm{Tm}\ A \vdash x \cdot z : \mathrm{Tm}\ (B \Rightarrow C)} \quad \dfrac{}{x : \mathrm{Tm}\ (A \Rightarrow B \Rightarrow C),\ y : \mathrm{Tm}\ (A \Rightarrow B),\ z : \mathrm{Tm}\ A \vdash y : \mathrm{Tm}\ (A \Rightarrow B)}\checkmark \quad \dfrac{}{x : \mathrm{Tm}\ (A \Rightarrow B \Rightarrow C),\ y : \mathrm{Tm}\ (A \Rightarrow B),\ z : \mathrm{Tm}\ A \vdash z : \mathrm{Tm}\ A}\checkmark}{x : \mathrm{Tm}\ (A \Rightarrow B \Rightarrow C),\ y : \mathrm{Tm}\ (A \Rightarrow B),\ z : \mathrm{Tm}\ A \vdash x \cdot z \cdot (y \cdot z) : \mathrm{Tm}\ C}}{\dfrac{x : \mathrm{Tm}\ (A \Rightarrow B \Rightarrow C),\ y : \mathrm{Tm}\ (A \Rightarrow B) \vdash \mathrm{lam}\ z. x \cdot z \cdot (y \cdot z) : \mathrm{Tm}\ (A \Rightarrow C)}{x : \mathrm{Tm}\ (A \Rightarrow B \Rightarrow C) \vdash \mathrm{lam}\ y. \mathrm{lam}\ z. x \cdot z \cdot (y \cdot z) : \mathrm{Tm}\ ((A \Rightarrow B) \Rightarrow A \Rightarrow C)}}{\mathrm{lam}\ x. \mathrm{lam}\ y. \mathrm{lam}\ z. x \cdot z \cdot (y \cdot z) : \mathrm{Tm}\ ((A \Rightarrow B \Rightarrow C) \Rightarrow (A \Rightarrow B) \Rightarrow A \Rightarrow C)}$$

`someTerm`

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{x : \mathrm{Tm}\ (A \Rightarrow Q),\ y : \mathrm{Tm}\ ((((P \Rightarrow Q) \Rightarrow P) \Rightarrow P) \Rightarrow A),\ z : \mathrm{Tm}\ ((P \Rightarrow Q) \Rightarrow P) \vdash x : \mathrm{Tm}\ (A \Rightarrow Q)}\checkmark \quad \dfrac{}{x : \mathrm{Tm}\ (A \Rightarrow Q),\ y : \mathrm{Tm}\ ((((P \Rightarrow Q) \Rightarrow P) \Rightarrow P) \Rightarrow A),\ z : \mathrm{Tm}\ ((P \Rightarrow Q) \Rightarrow P) \vdash z : \mathrm{Tm}\ ((P \Rightarrow Q) \Rightarrow P)}\checkmark}{x : \mathrm{Tm}\ (A \Rightarrow Q),\ y : \mathrm{Tm}\ ((((P \Rightarrow Q) \Rightarrow P) \Rightarrow P) \Rightarrow A),\ z : \mathrm{Tm}\ ((P \Rightarrow Q) \Rightarrow P) \vdash z \cdot (\mathrm{lam}\ w. x \cdot (y \cdot (\mathrm{lam}\ v. w))) : \mathrm{Tm}\ P} \quad \dfrac{}{x : \mathrm{Tm}\ (A \Rightarrow Q),\ y : \mathrm{Tm}\ ((((P \Rightarrow Q) \Rightarrow P) \Rightarrow P) \Rightarrow A),\ z : \mathrm{Tm}\ ((P \Rightarrow Q) \Rightarrow P) \vdash y : \mathrm{Tm}\ (\mathrm{lam}\ v. w) : \mathrm{Tm}\ Q}\checkmark}{x : \mathrm{Tm}\ (A \Rightarrow Q),\ y : \mathrm{Tm}\ ((((P \Rightarrow Q) \Rightarrow P) \Rightarrow P) \Rightarrow A),\ z : \mathrm{Tm}\ ((P \Rightarrow Q) \Rightarrow P) \vdash \mathrm{lam}\ w. x \cdot (y \cdot (\mathrm{lam}\ v. w)) : \mathrm{Tm}\ (P \Rightarrow Q)}\checkmark}{x : \mathrm{Tm}\ (A \Rightarrow Q),\ y : \mathrm{Tm}\ ((((P \Rightarrow Q) \Rightarrow P) \Rightarrow P) \Rightarrow A) \vdash \mathrm{lam}\ z. z \cdot (\mathrm{lam}\ w. x \cdot (y \cdot (\mathrm{lam}\ v. w))) : \mathrm{Tm}\ (((P \Rightarrow Q) \Rightarrow P) \Rightarrow P)}}{x : \mathrm{Tm}\ (A \Rightarrow Q),\ y : \mathrm{Tm}\ ((((P \Rightarrow Q) \Rightarrow P) \Rightarrow P) \Rightarrow A) \vdash y \cdot (\mathrm{lam}\ z. z \cdot (\mathrm{lam}\ w. x \cdot (y \cdot (\mathrm{lam}\ v. w)))) : \mathrm{Tm}\ A}}{x : \mathrm{Tm}\ (A \Rightarrow Q) \vdash \mathrm{lam}\ y. y \cdot (\mathrm{lam}\ z. z \cdot (\mathrm{lam}\ w. x \cdot (y \cdot (\mathrm{lam}\ v. w)))) : \mathrm{Tm}\ ((((P \Rightarrow Q) \Rightarrow P) \Rightarrow P) \Rightarrow A) \Rightarrow A)}}{\mathrm{lam}\ x. \mathrm{lam}\ y. y \cdot (\mathrm{lam}\ z. z \cdot (\mathrm{lam}\ w. x \cdot (y \cdot (\mathrm{lam}\ v. w)))) : \mathrm{Tm}\ ((A \Rightarrow Q) \Rightarrow ((((P \Rightarrow Q) \Rightarrow P) \Rightarrow P) \Rightarrow A) \Rightarrow A)}$$

Addendum

We may also add a function composition construct to the language, and use it to define `apply2x`, denoted $_ \circ _$. Note that an appropriate β rule needs to be defined for this new operator if we wish to prove equality between the two `apply2x` definitions.

$$\frac{t : \mathrm{Tm}\ (B \Rightarrow C) \qquad u : \mathrm{Tm}\ (A \Rightarrow B)}{t \circ u : \mathrm{Tm}\ (A \Rightarrow C)}$$

We derive the proof tree for `apply2x` once more, utilizing this new symbol and rule.

$$\dfrac{\dfrac{\dfrac{\checkmark}{f : \mathrm{Tm}\ (Nat \Rightarrow Nat) \vdash f : \mathrm{Tm}\ (Nat \Rightarrow Nat)} \quad \dfrac{\checkmark}{f : \mathrm{Tm}\ (Nat \Rightarrow Nat) \vdash f : \mathrm{Tm}\ (Nat \Rightarrow Nat)}}{f : \mathrm{Tm}\ (Nat \Rightarrow Nat) \vdash f \circ f : \mathrm{Tm}\ (Nat \Rightarrow Nat)}}{\mathrm{lam}\ \ f. f \circ f : \mathrm{Tm}\ ((Nat \Rightarrow Nat) \Rightarrow Nat \Rightarrow Nat)}$$